

1
2 Supplementary Information for *Interplay between climate*
3 *and childhood mixing can explain a sudden shift in RSV*
4 *seasonality in Japan*

5
6 Sang Woo Park^{1,2,3,*}, Inga Holmdahl^{3,4}, Emily Howerton³, Wenchang Yang⁵, Rachel
7 E. Baker⁶, Gabriel A. Vecchi^{4,5,7}, Sarah Cobey², C. Jessica E. Metcalf^{3,4,8}, Bryan T.
8 Grenfell^{3,4,8}

9 **1** School of Biological Sciences, Seoul National University, Seoul, Korea

10 **2** Department of Ecology and Evolution, University of Chicago, Chicago, IL, USA

11 **3** Department of Ecology and Evolutionary Biology, Princeton University,
12 Princeton, NJ, USA

13 **4** High Meadows Environmental Institute, Princeton University, Princeton, NJ,
14 USA

15 **5** Department of Geosciences, Princeton University, Princeton, NJ, USA

16 **6** Department of Epidemiology, Brown School of Public Health, Brown University,
17 Providence, Rhode Island, USA

18 **7** Program in Atmospheric and Oceanic Sciences, Princeton University, Princeton,
19 NJ, USA

20 **8** Princeton School of Public and International Affairs, Princeton, NJ, USA

21 *Corresponding author: sangwoopark@snu.ac.kr

22 Supplementary Materials

23 Supplementary Table

Island	Variable	Coefficient	p-value
Hokkaido	Intercept	4.28***	7.28×10^{-5}
	Humidity	1.26**	0.0011
	Humidity ²	-0.02	0.082
	Temperature	-0.13	0.051
	Temperature ²	-0.02***	7.75×10^{-6}
Honshu	Intercept	6.73***	5.4×10^{-13}
	Humidity	0.38	0.19
	Humidity ²	0.03*	0.011
	Temperature	0.03	0.80
	Temperature ²	-0.02***	0.00012
Shikoku	Intercept	12.25***	$< 2 \times 10^{-16}$
	Humidity	0.43	0.30
	Humidity ²	-0.01	0.67
	Temperature	-0.84***	0.00021
	Temperature ²	0.02*	0.013
Kyushu	Intercept	9.91***	$< 2 \times 10^{-16}$
	Humidity	0.52	0.068
	Humidity ²	0.01	0.27
	Temperature	-0.59***	0.00046
	Temperature ²	0.00	0.60
Ryukyu	Intercept	14.07	0.056
	Humidity	-0.90	0.060
	Humidity ²	0.04**	0.0073
	Temperature	0.15	0.86
	Temperature ²	-0.01	0.56

Table S1: **Bivariate quadratic regression of estimated transmission rate against mean specific humidity and mean temperature.** P-values were obtained from a two-sided t-test based on the estimated standard errors from the regression model. We did not adjust for multiple comparisons. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Island	Humidity model	Temperature model
Hokkaido	0.44	0.52
Honshu	0.39	0.29
Shikoku	0.63	0.72
Kyushu	0.77	0.77
Ryukyu	0.59	0.47

Table S2: Comparisons of R squared values from univariate quadratic regression of estimated transmission rate against mean specific humidity and mean temperature.

24 **Supplementary Figures**

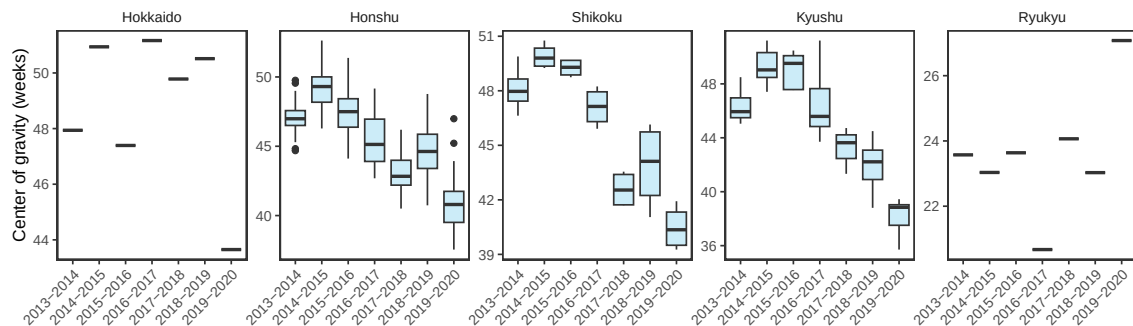


Figure S1: **Estimates of center of gravity (i.e., the mean timing of an epidemic) across all prefectures, stratified by island.** Hokkaido and Ryukyu islands each contain only one prefectures: Hokkaido and Okinawa, respectively. The center (horizontal line), lower bounds, and upper bounds of the box plot correspond to median, lower quartile (25th percentile), and upper quartile (75th percentile), respectively. The whiskers indicate the range of values that extend up to 1.5 times the interquartile range beyond the lower and upper quartiles. Outliers, which fall outside this range, are plotted individually.

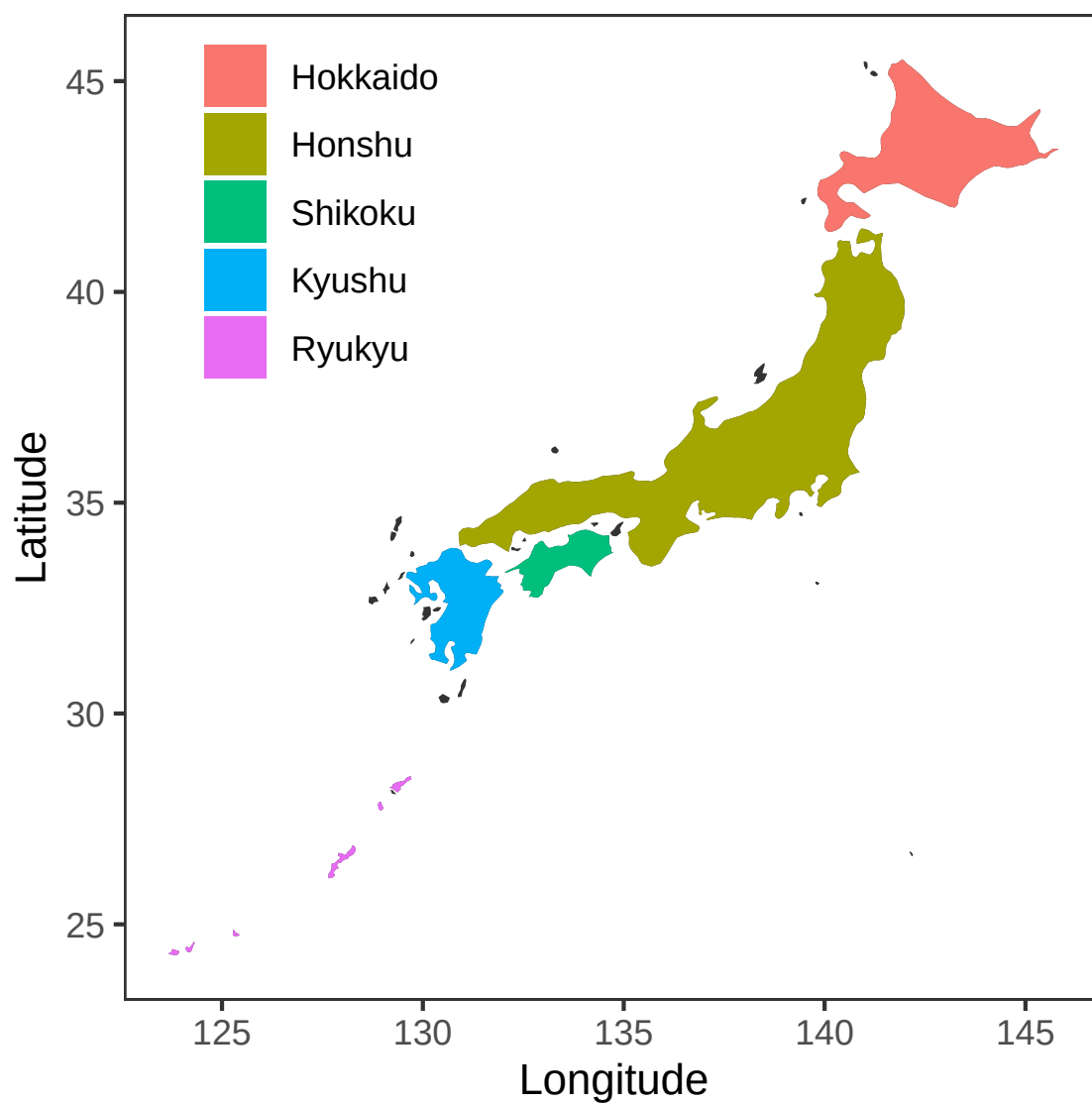


Figure S2: **Colored map of Japan.** Each of five major islands are marked by different colors.

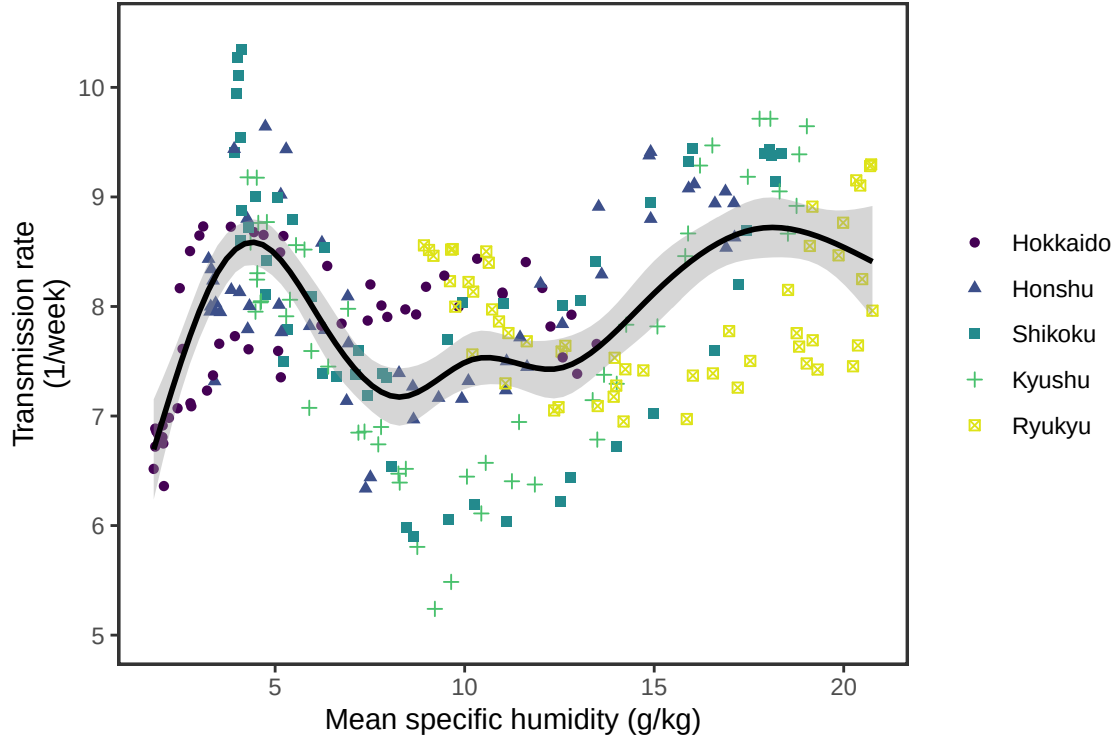


Figure S3: **Joint relationship between the estimated periodic seasonal transmission rates and mean specific humidity across all five islands.** Points represent the estimates across 52 weeks in each island. The line represent a generalized additive model fit using cubic spline basis. Shaded regions represent the corresponding 95% confidence intervals (n=260).

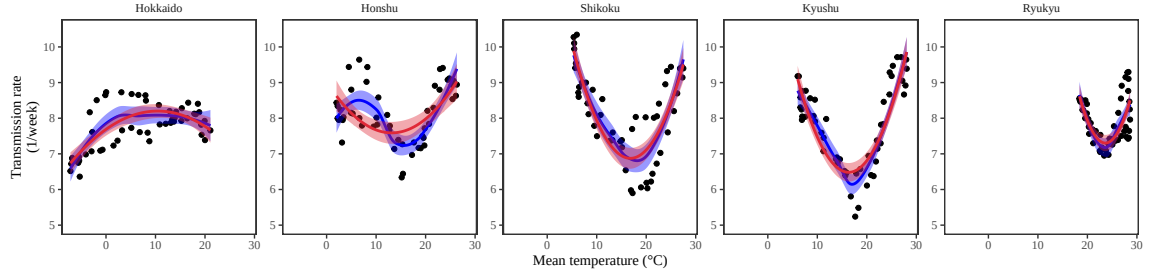


Figure S4: **Relationship between the estimated periodic seasonal transmission rates and mean temperature.** Points represent seasonal transmission rate estimates across 52 weeks versus average humidity across 2013–2020. Blue lines and regions represent the corresponding locally estimated scatterplot smoothing (LOESS) estimates and corresponding 95% confidence intervals ($n=52$). Red lines and regions represent the corresponding quadratic regression fits and corresponding 95% confidence intervals ($n=52$).

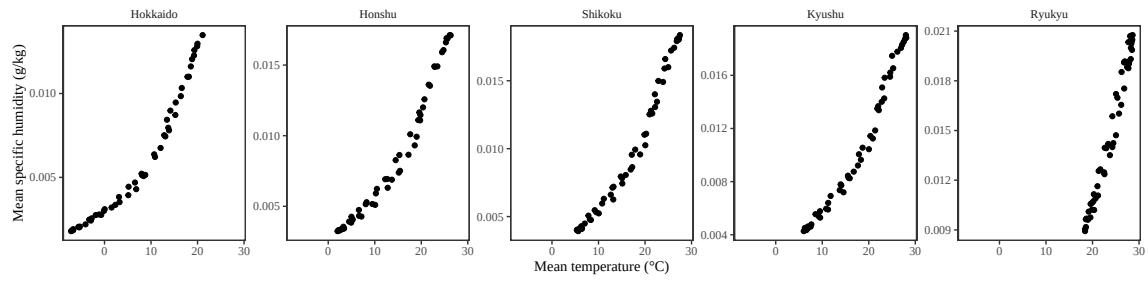


Figure S5: Relationship between the mean weekly temperature and mean weekly humidity.

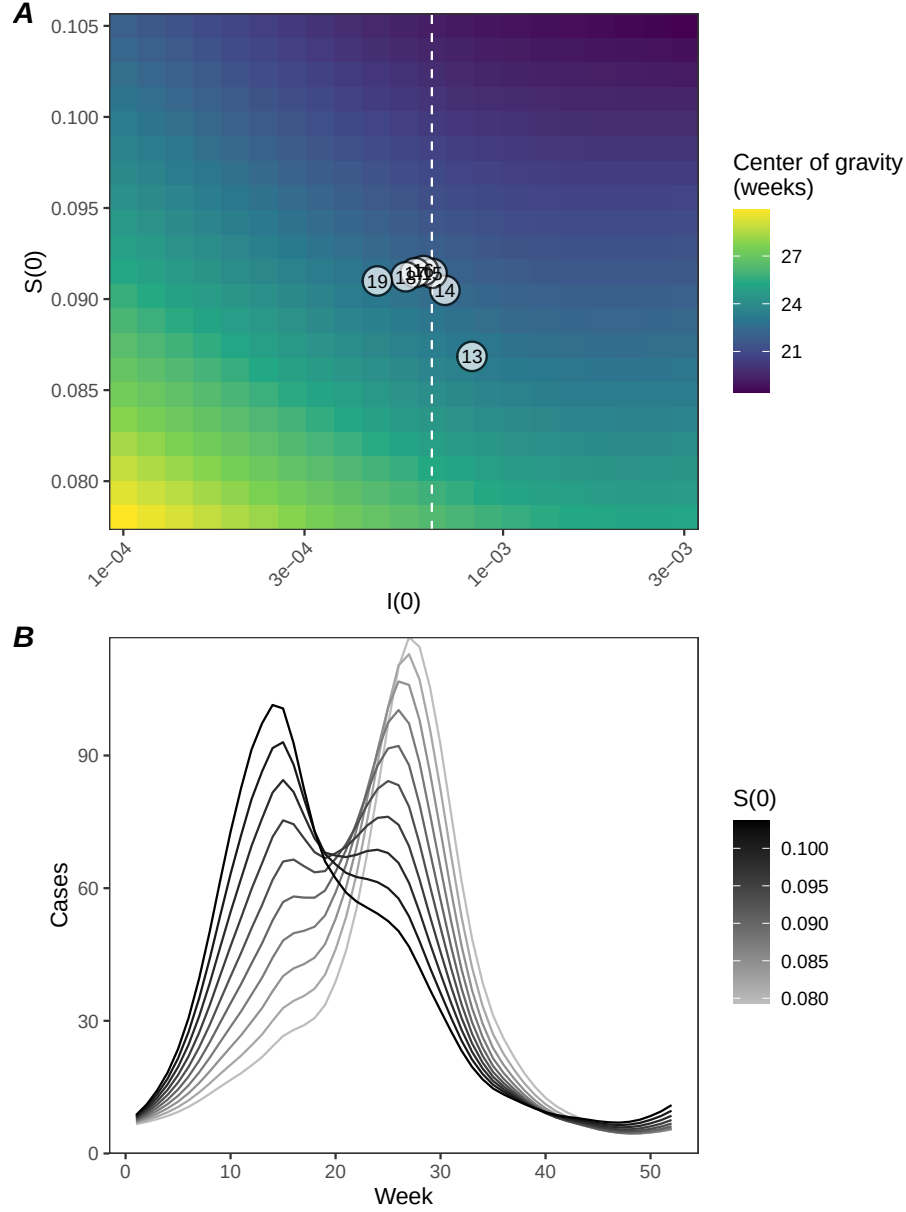


Figure S6: **A lack of increase in susceptible pool explains constant seasonality in Ryukyu island.** (A) Predicted effects of the proportion of infected $i(0)$ and susceptible $S(0)$ at the beginning of season on center of gravity. Points represent the estimated values for $i(0)$ and $s(0)$ between 2013 and 2019, showing the last two digits of a given year. The white vertical dashed line represents the $i(0)$ value used for simulating epidemic dynamics in panel B. (B) Changes in epidemic trajectories that would be caused by an increase in the susceptible proportion at the beginning of season for a fixed value of $i(0)$.

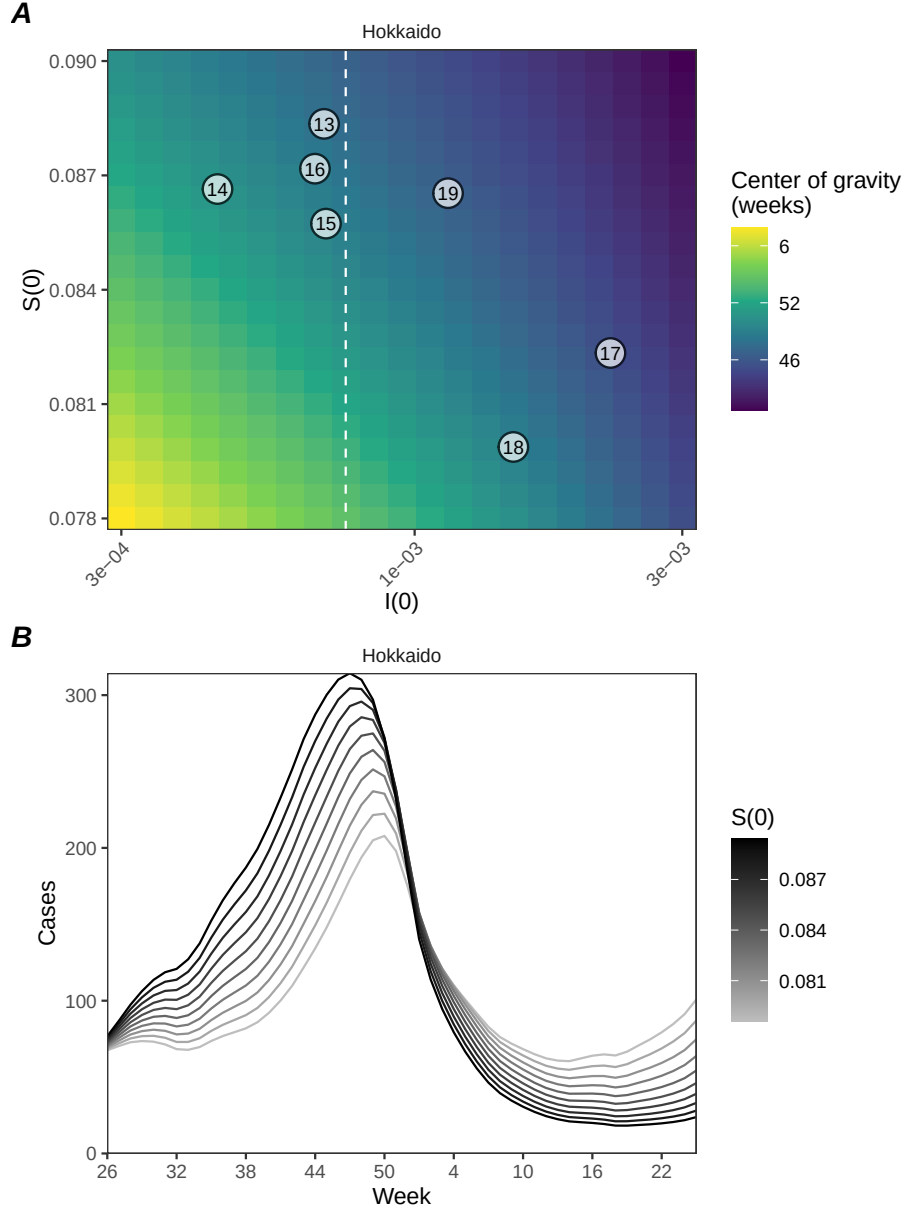


Figure S7: **Estimated change in susceptibility for Hokkaido island and counterfactual simulations assuming an increased susceptibility.** (A) Predicted effects of the proportion of infected $i(0)$ and susceptible $s(0)$ at the beginning of season on center of gravity. Points represent the estimated values for $i(0)$ and $s(0)$ between 2013 and 2019, showing the last two digits of a given year. The white vertical dashed line represents the $i(0)$ value used for simulating epidemic dynamics in panel B. (B) Changes in epidemic trajectories that would be caused by an increase in the susceptible proportion at the beginning of season for a fixed value of $i(0)$.

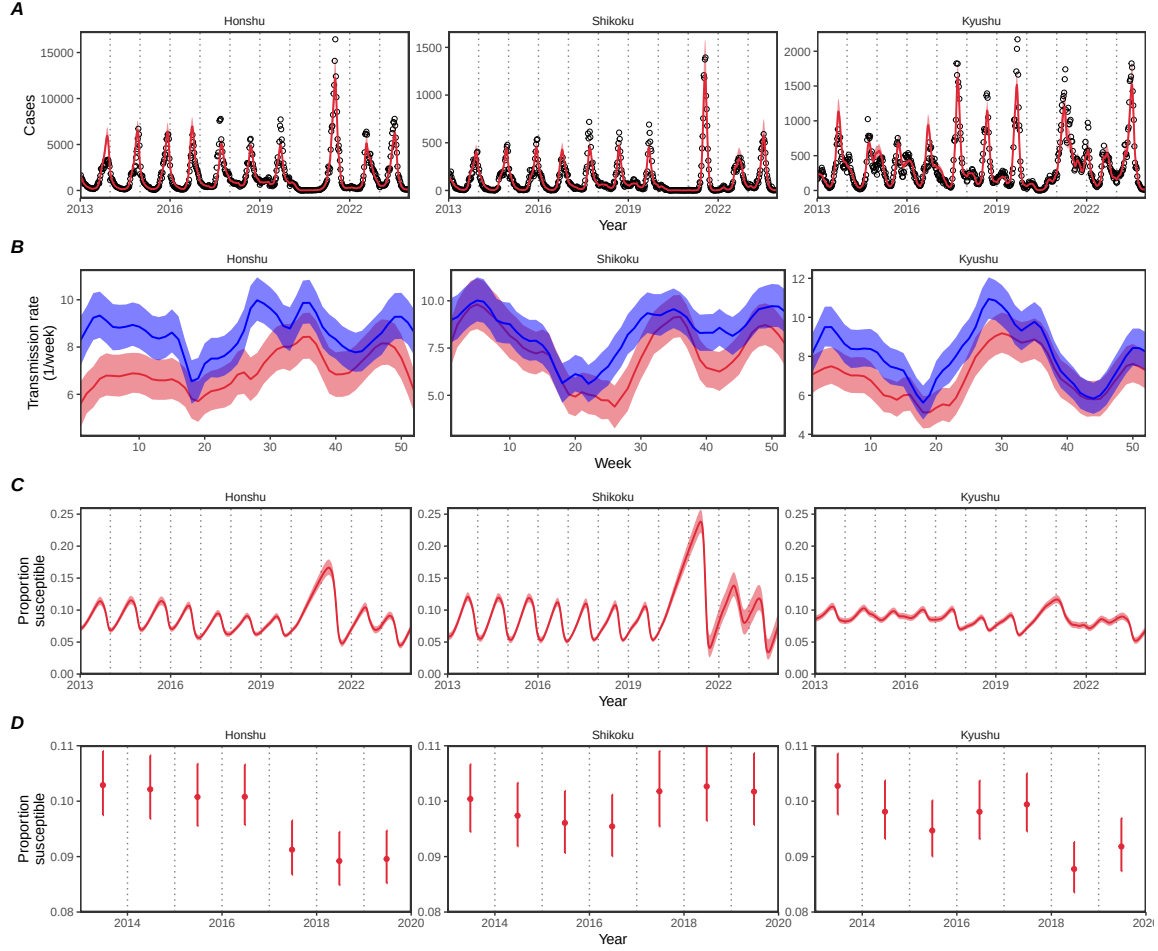


Figure S8: **Summary of SIRS model fits to RSV outbreaks across major islands in Japan.** (A) Comparisons of observed cases (points) across the five major islands and fitted epidemic trajectories (red lines). (B) Estimated periodic seasonal transmission rates before (red) and after (blue) the change in seasonality of RSV outbreaks. (C) Estimated proportion of the susceptible pool over time. In panels A–C, lines represent the estimated median of the posterior distribution ($n=8000$). In panels A–C, shaded regions represent the 95% credible intervals from the posterior distribution ($n=8000$). (D) Estimated proportion of the susceptible pool on the 26th week of each year. Points represent median from $n=8000$ posterior samples. Error bars represent the 95% credible interval in our estimates ($n=8000$ posterior samples).